MILITARY SPECIFICATION

FILTER, SIGNAL LIGHT MXU-507/E

1. SCOPE

1.1 This specification covers one type of visible light absorbing filter, designated MXU-507/E, for use on the SDU-5/E distress marker light.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-393	Plastic Molding Material, Polycarbonate, Injection and Extrusion
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Setup

Military

MIL-P-116	Preservation, Methods of
MIL-L-38217	Light, Marker Distress SDU-5/E

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods

DRAWINGS

Air Force

64A2067	Dome - Light, Marker, Distress SDU-5/E
64B2085	Dome, Base - Light, Marker, Distress SDU-5/E
64D2095	Case, Attachment - Light, Marker, Distress SDU-5/E, Assembly of

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

The Chemical Rubber Company

Handbook of Chemistry and Physics (47th Edition)

(Application for copies should be addressed to the Chemical Rubber Company, 2310 Superior Avenue, Cleveland, Ohio 44114.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials and components. Materials and components shall conform to applicable specifications and drawings as specified herein. Materials and components not covered by applicable specifications or not specifically described herein shall be of the best quality, of the lightest practicable weight, and entirely suitable for the purpose intended.

3.3.1 Protective treatment. If materials are used which are susceptible to deterioration when exposed to climatic and environmental conditions likely to occur during service under tropical conditions, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design. The filter shall be designed to fit snugly over the clear plastic dome (Drawing 64A2067) and extend over the lip formed by the lamp base (Drawing 64B2085) of the SDU-5/E distress marker light conforming to MIL-L-38217. The filter shall be designed to fit over lamp bases manufactured to either the minimum or maximum dimensions specified on Drawing 64B2085. When in place on the SDU-5/E light, the filter shall not extend beyond any part of the light assembly it covers by more than 3/16 inch, and it shall be possible to insert the entire assembly into and remove it from the carrying case conforming to Drawing 64D2095. The filter shall be notched on both ends

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to ensure a proper fit over the lanyard attachment loops on the SDU-5/E light. The dimensions of the notches shall be as shown on figure 1.

3.4.1 Sealing of gelatin filter between plastic sheets. If a gelatin-type filter is used to meet the requirements of this specification, it shall be sandwiched between plastic sheets conforming to L-P-393. The edges shall be sealed to protect the gelatin material from contaminants. All areas not covered by the filter material shall be opaque to visible light transmission (up to 7,200 angstroms).

3.4.2 Infrared transmitting portion. The minimum size of the infrared transmitting portions of the filter shall be in accordance with figure 1.

3.5 Performance

3.5.1 Transmission characteristics. The transmission characteristics of the assembled filter shall be within plus or minus 10 percent of the transmission characteristics of a Wratten Gelatin Filter 88A as covered in the Chemical Rubber Company Handbook of Chemistry and Physics.

3.5.2 Passage of visual light. The filter shall fit firmly in place over the plastic dome of the SDU-5/E light so that when the light is in operation (below 7,200 angstroms), visible light shall not pass through the filter or between the mating edges of the filter and the light.

3 Environmental conditions

3.5.3.1 High temperature and humidity. The filter shall be capable of withstanding exposure to the following conditions without the light transmission characteristics being degraded by more than 20 percent of the values established in 3.5.1, and in no case shall there be light transmission below 7,200 angstroms:

a. Temperature of 122°F

b. Relative humidity of 95 percent.

3.5.3.2 Fungus. The filter shall not support the growth of fungus.

3.5.3.3 Low temperature exposure. The filter shall withstand a temperature of -65°F plus or minus 2°F for a period of 12 hours without distortion or damage which would affect the light transmission characteristics as established in 3.5.1.

3.5.4 Separation force. A force of 4 pounds (plus or minus 1 pound) shall be required to remove the filter from the light assembly.

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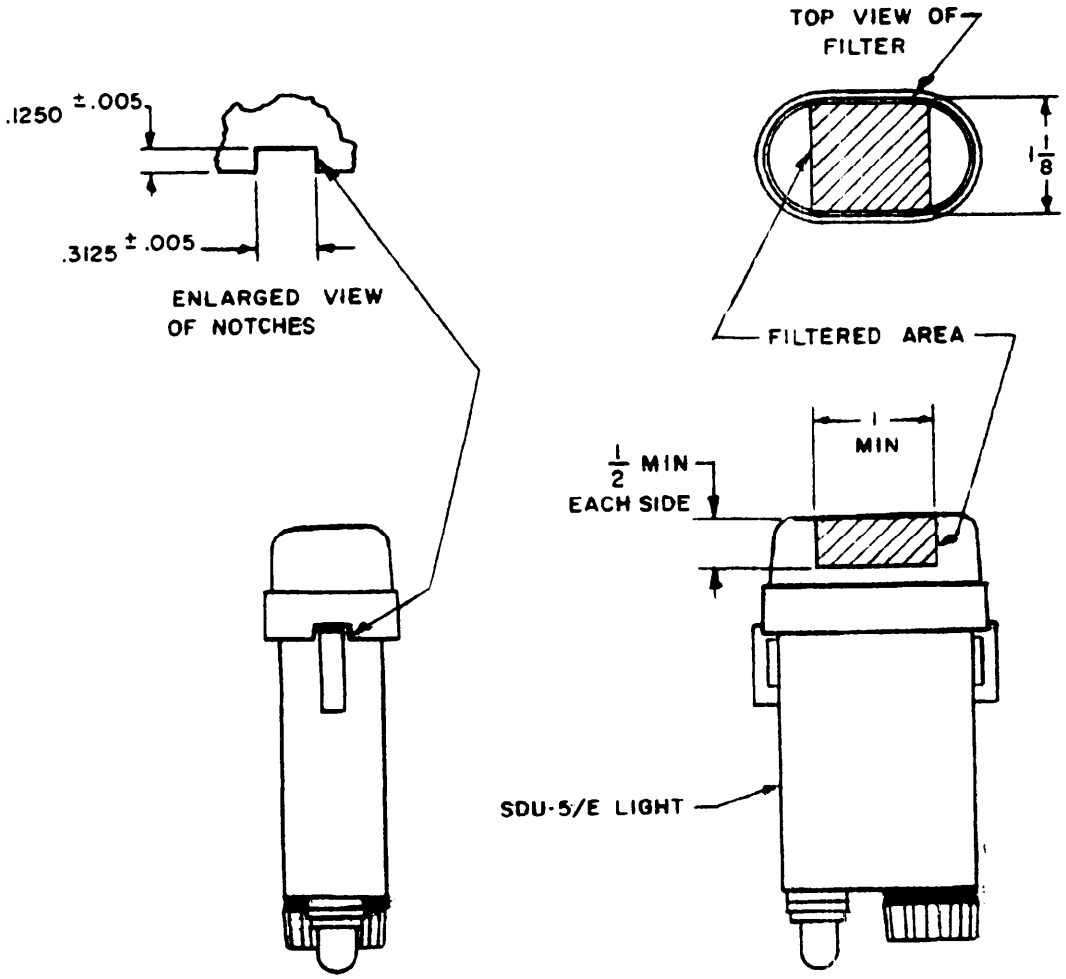


FIGURE 1. Filter, Signal Light MXU-507/E

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3.5.5 Compression. It shall be possible to compress the sides of the assembled filter by one-half of the nominal dimension and the ends by one-eighth of the dimension without causing cracks or otherwise degrading the performance of the filter.

3.6 Government-loaned property. When specified in the contract (see 6.2), the Government will loan the following to the contractor:

Item No.	Stock No.	Item Description	Quantity
1	6230-067-5209	Light, Marker Distress SDU-5/E (MIL-L-38217) with Case, Attachment - Light Marker, Distress SDU-5/E (Drawing 64D2095)	2 each

3.7 Workmanship. The filter shall be constructed in accordance with high grade manufacturing practices covering this type of material. Particular attention shall be given to freedom from paths (breaks, scratches, cracks, etc).

4. QUALITY ASSURANCE PROVISIONS

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the filter shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77°F (plus or minus 18°F), and at a relative humidity of 80 percent or less.

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4.4 Preproduction testing

4.4.1 Preproduction test samples. The contractor shall subject six filters to the preproduction tests as specified in 4.4.2. The six filters selected may be used for all tests; however, an additional six filters may be used to conduct the tests specified in 4.6.5 and 4.6.6.

4.4.2 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6. The number of filters to be subjected to each test and the order for conducting the tests shall be as follows:

Test	Number of Filters to be Tested
4.6.1	6
4.6.2	6
4.6.3	6
4.6.4.1	2
4.6.4.2	2
4.6.4.3	2
4.6.4.4	1
4.6.5	6
4.6.6	6

4.4.3 Preproduction test report. When specified (see 6.2), a preproduction test report shall be prepared.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each filter shall be subjected to the examination of product test described in 4.6.1.

4.5.2 Sampling tests. Five filters shall be selected at random from each lot of 500, or fraction thereof, produced under essentially the same conditions

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and submitted for inspection at substantially the same time. The test samples shall be subjected to the following tests:

a. Transmission characteristics (4.6.2)

b. Passage of visual light (4.6.3)

c. Separation force (4.6.5).

4.5.2.1 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of the failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The filter shall be examined to determine conformance to the requirements specified herein with respect to workmanship, materials, and dimensions. The filter shall be examined with a light of sufficient intensity to determine that no paths (breaks, scratches, cracks, etc) exist which will permit visual light to pass.

4.6.2 Transmission characteristics. The spectrophotometric characteristics shall be measured to determine that the filter falls within and meets the limits specified in 3.5.1. The data obtained shall also be used as baselines for other tests (4.6.4.1 and 4.6.4.2).

4.6.3 Passage of visual light. The filter shall be installed on an SDU-5/E light. The light shall then be operated in a room where no visual light is present. The filter shall meet the requirements as specified in 3.5.2.

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4.6.4 Environmental tests

4.6.4.1 High temperature exposure. The filter shall be placed in a test chamber and the chamber temperature raised to 122°F and maintained for a period of not less than 48 hours. The internal chamber relative humidity shall not exceed 15 percent. At the conclusion of the exposure period, the filter shall be gradually returned to room conditions and tested in accordance with 4.6.2. The test data shall be compared with similar test data obtained for the particular filter during the original test specified in 4.6.2 to determine compliance with 3.5.3.1.

4.6.4.2 Humidity. The filter shall be subjected to a humidity test in accordance with method 507, procedure V of MIL-STD-810 except that the maximum temperature shall not exceed 122°F . At the conclusion of the test period, the filter shall be subjected to the test in 4.6.2. The results shall be compared with the results obtained from the original test specified in 4.6.2 to determine compliance with 3.5.3.1.

4.6.4.3 Fungus. The filter shall be tested in accordance with method 508, procedure I of MIL-STD-810.

4.6.4.4 Low temperature exposure. A filter shall be placed in a test chamber and the chamber temperature lowered to -65°F plus or minus 2°F for a period of 12 hours. The filter shall be removed after the 12-hour period and immediately subjected to the test specified in 4.6.5.

4.6.5 Separation force. An SDU-5/E light, with the filter in place, shall be attached in a fixed position to a test device, and a scale or device capable of measuring the force required to separate the filter from the light shall be attached to the filter. The filter shall then be separated from the light and the force required to effect removal shall be measured to determine compliance with 3.5.4.

4.6.6 Compression. The filter shall be placed in a clamp or device which can be used to compress the ends and then the sides of the filter so that the amount of compression can be measured to determine compliance with 3.5.5. Upon completion of this test, the filter shall be re-examined with a light source to determine it meets the test specified in 4.6.1.

4.7 Inspection of preparation for delivery requirements. Sample items and packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 or the documents specified therein.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Filters shall be packaged one each, method III, of MIL-P-116. Unit containers shall conform to PPP-B-566 or PPP-B-676.

5.1.2 Level C. Filters shall be packaged in a manner which will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. A quantity of filters packaged as specified in 5.1 shall be packed in fiberboard shipping containers conforming to PPP-B-636, class weather resistant. Insofar as practicable, shipping containers shall be uniform in shape and size and contain identical quantities. Container closure, sealing, and strapping shall be in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Level B shall be the same as level A except fiberboard shipping containers shall conform to class domestic.

5.2.3 Level C. A quantity of filters packaged as specified in 5.1 shall be packed in a manner which affords adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the suppliers commercial practice when such meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The filter covered by this specification is intended to cover the lens of the SDU-5/E distress marker light to absorb the visible spectrum from the light and transmit only the infrared spectrum.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Government-loaned property (see 3.6)

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c. Data requirements (see 4.4.3)

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

Custodian:
Air Force - 11

Review activity:
Air Force - 82

Preparing activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8274

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